THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 9: **Cell: The Building Block of Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The smooth ER in liver cells helps in: 1

(a) digestion of starch **(b)** cell division
(c) detoxification **(d)** photosynthesis

2. The structure that separates the nucleus from the cytoplasm is the: 1

(a) nuclear membrane **(b)** cell membrane **(c)** vacuole membrane **(d)** cell wall

3. A cell with a lot of mitochondria is most likely to be: 1

(a) a dead cell **(b)** a muscle cell **(c)** a fat cell **(d)** a cork cell

4. Potato cubes placed in strong salt solution: 1

(a) gain mass **(b)** lose mass **(c)** dissolve **(d)** stay same

5. The structure that is not part of the endomembrane system: 1

(a) ER **(b)** mitochondria **(c)** lysosome **(d)** Golgi

6. The nuclear pores allow: 1

(a) exchange between nucleus and cytoplasm **(b)** respiration
(c) photosynthesis **(d)** division

7. A cell with abundant rough ER and Golgi is likely to be: 1

(a) a leaf cell **(b)** a red blood cell
(c) a gland cell secreting proteins **(d)** a bone cell

8. Potato cubes placed in distilled water: 1

(a) stay same **(b)** lose mass **(c)** gain mass **(d)** burn

9. A student sees a cell with a large vacuole, cell wall and chloroplasts. The cell is from a: 1

(a) leaf **(b)** animal **(c)** bacterium **(d)** root tip

10. Chromatin condenses to form: 1

(a) ribosomes **(b)** vacuoles **(c)** plastids **(d)** chromosomes

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. How do you show that the cell membrane is selectively permeable using beetroot? 2

12. Why does a plant cell not burst in pure water? 2

13. Why do salted fruits and meat last longer? 2

14. Why does a drop of ink spread faster in hot water than in cold? Relate to diffusion. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the structure of mitochondria is related to its function. 3

16. Describe the changes in a plant cell placed in sugar solution of increasing concentration. 3

17. Explain osmosis in terms of water potential or concentration, using a semipermeable membrane and two sugar solutions. 3

OR

Explain the process of formation of a vesicle from the ER to the Golgi and the release of a protein from a cell. 3

18. State three differences between mitochondria and chloroplasts. 3

OR

List any three precautions in the osmosis experiment with potato and state the conclusion. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the structure, function and importance of plastids in plant cells with the types and their examples. 5

OR

A plant cell and an animal cell are placed in distilled water. Describe the changes, the reason for the difference and the role of the contractile vacuole in Amoeba. 5

20. Compare plant, animal and bacterial cells in a table with organelles and features. 5

OR

Describe how scientists study cells: microscopes, stains, cell fractionation, cell culture, and mention two discoveries. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND